THE AREA FUNCTION OF THE MERTENS EXCURSION SKELETON (RESEARCH NOTEBOOK, VERSION 8 — AUDITED MERGED EDITION)

V. CASAÑA-GINER

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Remark 0.1. This notebook is exploratory. Its purpose is to study the Mertens function through its excursions between zero runs, and to separate what is forced by the local geometry of those excursions from what requires arithmetic input. It contains exact identities, conditional statements, heuristics and archived statements from earlier versions. Every statement is to be read with an explicit evidentiary status. Version 8 distinguishes: *THEOREM / PROVED*; *PROVED BUT WEAK*; *NO USEFUL BOUND KNOWN FROM THE PRESENT ARGUMENT*; *CONJECTURE / HEURISTIC*; and *FALSE / WITHDRAWN*. Earlier statements that are incorrect or superseded are retained only when their failure is itself informative, and are marked as such.

1. Introduction and conventions

The main object is the Mertens function M, decomposed into excursions between zero runs, together with three area curves built from a regularized skeleton of those excursions (Section 2). Sections 3 to 7 record exact formulas, sharp bounds and criteria equivalent to the Riemann hypothesis. Section 8 contains the two new lacunarity bounds for long excursions and the resulting

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localized RH criteria, including the reduction to excursions ending before x_0^2. Sections 9 to 11 then examine the square trial-division horizon and large-prime decompositions.

Audit convention. The following labels are literal. *[THEOREM]* means proved unconditionally, either here or by the cited literature. *[PROVED BUT WEAK]* means rigorous but insufficient for the stated RH objective. *[UNDER RH]* or another named hypothesis means conditional. *[NO COTA]* means that the present argument supplies no non-trivial bound of the required type. *[CONJECTURE]* and *[HEURISTIC]* are unproved. *[FALSE / WITHDRAWN]* identifies an earlier inference that is not valid. An unconditional equivalence with RH is a theorem and does not assume RH.

Status lines. Each section opens with a status line that separates standard material, literature-based material, original observations, heuristics and conjectures, as required by the authoring rules of this notebook.

Notation changes with respect to version 5. The symbol $m(x)$ of version 5 is replaced by $M(x)$. The unregularized skeleton, called L in version 5, is denoted M_{sk}. The squarefree counting function, called $Q(y)$ in version 5, is denoted $\mathcal{S}(y)$, since Q_ν denotes the quadratic energy of an excursion. The Dirichlet energy, called $E(x)$ in version 5, is denoted $\mathcal{E}(x)$. The coefficient called $A(R)$ in version 5 is denoted $\Theta(R)$, since $\mathcal{A}$ denotes the black curve. The ratio t/u^2 keeps the name R; the residual $M - M_{\mathrm{sk}}$ of version 5, also called R there, is written out explicitly. The defect parameter of Remark 7.12 is denoted d_ν.

2. Excursions and the regularized skeleton

Status. The segmentation of M by zero runs and the record of its extrema are standard computational objects; Hurst [7] computed $M(x)$ for all $x \le 10^{16}$, recording all zeros and extrema. The area and quadratic functionals of lattice excursions are classical [12, 10, 11, 8, 2, 1]. What is specific to this notebook is the regularized triangular skeleton and its use on the deterministic Möbius walk. All statements in this section are unconditional.

Let μ be the Möbius function and

$$M(x) = \sum_{n \le x} \mu(n), \qquad x \ge 0, \tag{2.1}$$

the Mertens function, piecewise constant and right-continuous, with jumps $\mu(n) \in \{-1, 0, 1\}$ at the integers.

Definition 2.1 (Excursions and zero runs)**.** An *excursion* is an interval

$$I_\nu = (x_{0,\nu}, x_{1,\nu}] = (u_\nu, v_\nu]$$

with $M(x_{0,\nu}) = M(x_{1,\nu}) = 0$, $M(x_{0,\nu}+1) = \sigma_\nu \in \{\pm 1\}$, and $\sigma_\nu M(n) \ge 1$ for every integer $x_{0,\nu} < n < x_{1,\nu}$. Thus $x_{1,\nu}$ is the first return to zero after the departure from $x_{0,\nu}$. Consecutive excursions may be separated by a *zero run*

$$[x_{1,\nu}, x_{0,\nu+1}],$$

a maximal set of consecutive integers on which $M = 0$; it may consist of a single point. Zero runs are not excursions. The first excursion is $(0, 2]$. We retain u_ν, v_ν as legacy aliases for $x_{0,\nu}, x_{1,\nu}$ in formulas inherited from versions 5–6.

Put

$$\ell_\nu = x_{1,\nu} - x_{0,\nu}, \qquad H_\nu = \max_{x_{0,\nu} < n \le x_{1,\nu}} |M(n)|,$$

$$\tau_\nu = \min\{n \in I_\nu : |M(n)| = H_\nu\}, \qquad a_\nu = \tau_\nu - x_{0,\nu}, \qquad b_\nu = x_{1,\nu} - \tau_\nu.$$

Every excursion has $H_\nu \ge 1$ and $\ell_\nu \ge 2$.

Lemma 2.2 (Height and step count)**.** *For every excursion,*

$$H_\nu \le \min(a_\nu, b_\nu) \le \frac{\ell_\nu}{2}. \tag{2.2}$$

Proof. The steps of M are 0 or ± 1, so at least H_ν steps are needed to reach height H_ν from zero and at least H_ν more to return to zero. □

Definition 2.3 (Crossing and touching zero runs)**.** The zero run $[x_{1,\nu}, x_{0,\nu+1}]$ between consecutive excursions is a *crossing run* if $\sigma_{\nu+1} = -\sigma_\nu$ and a *touching run* if $\sigma_{\nu+1} = \sigma_\nu$.

Definition 2.4 (Regularized skeleton)**.** The *skeleton* M_{s} is the continuous, piecewise linear function defined on every excursion by

$$M_{\mathrm{s}}(t) = \begin{cases} \sigma_\nu \sqrt{H_\nu}\, \dfrac{t - x_{0,\nu}}{a_\nu}, & x_{0,\nu} \le t \le \tau_\nu, \\[2ex] \sigma_\nu \sqrt{H_\nu}\, \dfrac{x_{1,\nu} - t}{b_\nu}, & \tau_\nu \le t \le x_{1,\nu}, \end{cases} \tag{2.3}$$

and by $M_{\mathrm{s}} \equiv 0$ on zero runs. The *unregularized skeleton* M_{sk} is defined in the same way with peak $\sigma_\nu H_\nu$.

Definition 2.5 (Terminal block of a truncated computation)**.** Let $N \ge 2$ and let $u = \max\{n \le N : M(n) = 0\}$. If $u = N$ there is no terminal block. Otherwise put

$$\sigma_{\mathrm{term}} = \operatorname{sgn} M(N), \qquad H_{\mathrm{term}} = \max_{u < n \le N} |M(n)|,$$

$$\tau_{\mathrm{term}} = \min\{n \in (u, N] : |M(n)| = H_{\mathrm{term}}\}.$$

On $[u, N]$ the regularized skeleton is the segment from $(u, 0)$ to $(\tau_{\mathrm{term}}, \sigma_{\mathrm{term}}\sqrt{H_{\mathrm{term}}})$ followed, if $\tau_{\mathrm{term}} < N$, by the segment ending at $(N, \sigma_{\mathrm{term}}\sqrt{|M(N)|})$. Its signed area is

$$\Delta\mathcal{A}(u, N] = \frac{\sigma_{\mathrm{term}}}{2}\Big[\sqrt{H_{\mathrm{term}}}(N - u) + \sqrt{|M(N)|}(N - \tau_{\mathrm{term}})\Big].$$

The terminal skeleton is provisional: after the next return to zero the completed excursion acquires a different descending side.

Definition 2.6 (The three curves)**.** With $M_{\mathrm{s}}^{\pm} = \max(\pm M_{\mathrm{s}}, 0)$,

$$\mathcal{A}_+(x) = \int_0^x M_{\mathrm{s}}^+(t)\, dt, \qquad \mathcal{A}_-(x) = -\int_0^x M_{\mathrm{s}}^-(t)\, dt, \qquad \mathcal{A}(x) = \int_0^x M_{\mathrm{s}}(t)\, dt. \tag{2.4}$$

$\mathcal{A}_+$ is nondecreasing, $\mathcal{A}_-$ is nonincreasing, and both are constant on zero runs and on excursions of the opposite sign.

Remark 2.7 (The family of g-skeletons). For any $g : \mathbb{N}_0 \to [0, \infty)$ with $g(0) = 0$ and $g(h) > 0$ for $h \ge 1$, one may replace the apex height $\sqrt{H}$ by $g(H)$. The signed area increment becomes $\frac{1}{2}\sigma_\nu g(H_\nu)\ell_\nu$. The correlation identities below do not depend on g. The choice $g(H) = \sqrt{H}$ is a normalization, not a geometric necessity.

3. Exact formulas

Status. Exact elementary formulas with complete proofs. The critical-geometry lemma includes the correction for positive-length zero plateaus.

Lemma 3.1 (Increment per excursion)**.** *For every complete excursion,*

$$\mathcal{A}(x_{1,\nu}) - \mathcal{A}(x_{0,\nu}) = \sigma_\nu \frac{\sqrt{H_\nu}\, \ell_\nu}{2}. \tag{3.1}$$

The discrete sum agrees:

$$\sum_{n\in I_\nu} M_{\mathrm{s}}(n) = \sigma_\nu \sqrt{H_\nu}\frac{\ell_\nu}{2}. \tag{3.2}$$

Proof. The two pieces are triangles of height $\sqrt{H_\nu}$ and bases a_ν, b_ν. The discrete ascent contributes $\sqrt{H_\nu}(a_\nu+1)/2$ and the descent $\sqrt{H_\nu}(b_\nu-1)/2$. □

Proposition 3.2 (Explicit form of the three curves)**.** *Let $x \in I_\nu$. Then*

$$\mathcal{A}(x) = \begin{cases} \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\dfrac{(x-x_{0,\nu})^2}{2a_\nu}, & x \le \tau_\nu, \\ \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\left(\dfrac{a_\nu}{2} + (x-\tau_\nu) - \dfrac{(x-\tau_\nu)^2}{2b_\nu}\right), & x \ge \tau_\nu. \end{cases} \tag{3.3}$$

At excursion endpoints,

$$\mathcal{A}(x_{1,k}) = \sum_{\nu\le k} \sigma_\nu \frac{\sqrt{H_\nu}\ell_\nu}{2}, \tag{3.4}$$

$$\mathcal{A}_+(x_{1,k}) = \sum_{\substack{\nu\le k \\ \sigma_\nu=+1}} \frac{\sqrt{H_\nu}\ell_\nu}{2}, \tag{3.5}$$

$$\mathcal{A}_-(x_{1,k}) = -\sum_{\substack{\nu\le k \\ \sigma_\nu=-1}} \frac{\sqrt{H_\nu}\ell_\nu}{2}. \tag{3.6}$$

The curves are piecewise quadratic and C^1.

Proof. Integrate (2.3); the endpoint formulas follow from Lemma 3.1. □

Lemma 3.3 (Critical geometry of $\mathcal{A}$)**.** *Let $R = [p, q]$ be the zero run between an excursion of sign σ_- ending at p and the next excursion of sign σ_+ starting at q.*

(i) *Every apex τ_ν is a strict inflection point.*
(ii) *If $\sigma_+ = -\sigma_-$, then R is an extremal plateau; it is a strict local extremum exactly when $p = q$.*
(iii) *If $\sigma_+ = \sigma_- =: \sigma$, then $\mathcal{A}$ has no local extremum on R. Its concavity has sign $-\sigma$ immediately to the left and $+\sigma$ immediately to the right. If $p = q$, the point is a strict inflection; if $p < q$, no point of R is a strict inflection.*
(iv) *Strict local extrema occur exactly at isolated crossing zeros, and plateaus of extrema exactly at crossing runs of positive length.*

Proof. On the descending side of the left excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_-$ and $\operatorname{sgn}\mathcal{A}'' = -\sigma_-$. On the ascending side of the right excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_+$ and $\operatorname{sgn}\mathcal{A}'' = \sigma_+$. On R, $\mathcal{A}' = 0$. The assertions follow, and at every apex the sign of $\mathcal{A}''$ reverses. □

Remark 3.4 (Status of the version 5 statement). Version 5 stated that every touching zero is an inflection point. This is correct verbatim only for a touching run consisting of one zero. On a positive-length touching run, $\mathcal{A}''$ has the pattern $-\sigma, 0, +\sigma$ and the change of concavity is carried by the whole plateau.

4. Internal correlations of an excursion

Status. Exact elementary identities with complete proofs. The triangular identity holds for arbitrary real sequences; the fixed-length upper bound for Q is sharp with leading constant $1/12$, correcting the constant $1/4$ in version 6.

For a real block $x_1, \dots, x_L$, put

$$S_t = \sum_{j=1}^{t} x_j, \qquad C_h = \sum_{j=1}^{L-h} x_j x_{j+h}. \tag{4.1}$$

Lemma 4.1 (Triangular identity)**.** *For every real finite sequence,*

$$\sum_{t=1}^{L-1} S_t^2 + \sum_{h=1}^{L-1} hC_h = S_L \sum_{t=1}^{L-1} S_t. \tag{4.2}$$

Proof. Since $\sum_{t=1}^{L} S_t^2 = \sum_{i,j} x_i x_j (L+1-\max(i,j))$, use $\max(i,j) = \frac{1}{2}(i+j+|i-j|)$ and $\sum_i i x_i = LS_L - \sum_{t=1}^{L-1} S_t$, then subtract S_L^2. □

Corollary 4.2 (Bridge case)**.** *If $S_L = 0$, then*

$$\sum_{t=1}^{L-1} S_t^2 = -\sum_{h=1}^{L-1} hC_h. \tag{4.3}$$

Remark 4.3 (Hermitian form). For complex x_j, with $C_h = \sum_j x_j \overline{x_{j+h}}$,

$$\sum_{t=1}^{L-1} |S_t|^2 + \sum_{h=1}^{L-1} h\Re C_h = \Re\left(\overline{S_L} \sum_{t=1}^{L-1} S_t\right).$$

For a complete excursion, orient the increments by

$$x_j = \sigma_\nu \mu(x_{0,\nu} + j), \qquad S_t = |M(x_{0,\nu} + t)|. \tag{4.4}$$

Then $S_t > 0$ for $1 \le t < \ell_\nu$ and $S_{\ell_\nu} = 0$. Put

$$Q_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t^2, \qquad C_h^{(\nu)} = \sum_{j=1}^{\ell_\nu - h} x_j x_{j+h}. \tag{4.5}$$

Thus

$$Q_\nu = -\sum_{h=1}^{\ell_\nu - 1} hC_h^{(\nu)}. \tag{4.6}$$

Lemma 4.4 (Zeroth moment)**.** *If $q_\nu = \sum_j x_j^2$, then*

$$\sum_{h=1}^{\ell_\nu - 1} C_h^{(\nu)} = -\frac{q_\nu}{2}. \tag{4.7}$$

Proof. Expand $(\sum_j x_j)^2 = 0$. □

Lemma 4.5 (Extreme lag)**.** *For every complete excursion,*

$$C_{\ell_\nu - 1}^{(\nu)} = -1,$$

and q_ν is even and at least 2*.*

Proof. The first oriented step is $+1$. Immediately before the first return to zero the path is at height 1, so the last step is -1. The bridge condition forces equally many positive and negative active steps. □

Proposition 4.6 (Quadratic decomposition by excursions)**.** *For real $x \ge 1$, let*

$$u(x) = \max\{n \le \lfloor x \rfloor : M(n) = 0\}, \qquad L_x = \lfloor x \rfloor - u(x).$$

For $1 \le j \le L_x$ put $y_j = \mu(u(x)+j)$, $S_t = M(u(x)+t)$ and $C_h^{\mathrm{term}} = \sum_{j=1}^{L_x-h} y_j y_{j+h}$. Then

$$\sum_{n\le x} M(n)^2 = \sum_{\nu: x_{1,\nu}\le u(x)} Q_\nu + T(x), \tag{4.8}$$

where

$$T(x) = M(\lfloor x\rfloor)^2 + M(\lfloor x\rfloor)\sum_{t=1}^{L_x-1} M(u(x)+t) - \sum_{h=1}^{L_x-1} hC_h^{\mathrm{term}}. \tag{4.9}$$

If $L_x = 0$, $T(x) = 0$. At a zero x,

$$\sum_{n\le x} M(n)^2 = \sum_{x_{1,\nu}\le x} Q_\nu = -\sum_{x_{1,\nu}\le x}\sum_{h} hC_h^{(\nu)}. \tag{4.10}$$

Proof. Every integer up to $u(x)$ lies either in a zero run or in a unique excursion interior. Apply Lemma 4.1 to the terminal block and add its final $S_{L_x}^2$. □

Remark 4.7 (Blindness to pairs crossing excursion boundaries). Globally,

$$\sum_{n\le x} M(n)^2 = \sum_{a,b\le x} \mu(a)\mu(b)(x+1-\max(a,b)).$$

Pairs crossing a complete excursion cancel because $\sum_{n\in I_\nu}\mu(n) = 0$; Proposition 4.6 retains only within-excursion pairs and the terminal block.

Let

$$\beta_\nu = \sum_{t=1}^{\ell_\nu-1} S_t = \sum_{n\in I_\nu} |M(n)|. \tag{4.11}$$

Cauchy–Schwarz gives the sharp factor

$$\beta_\nu^2 \le (\ell_\nu - 1)Q_\nu, \tag{4.12}$$

with equality exactly for a flat height-1 excursion.

To separate support from sign geometry, let $P_h^{(\nu)} = \#\{j : x_j x_{j+h} \ne 0\}$ and let $D_h^{(\nu)}$ count the opposite-sign pairs among them. Then

$$C_h^{(\nu)} = P_h^{(\nu)} - 2D_h^{(\nu)}, \tag{4.13}$$

and

$$Q_\nu = \sum_{h=1}^{\ell_\nu-1} h(2D_h^{(\nu)} - P_h^{(\nu)}). \tag{4.14}$$

Proposition 4.8 (Sharp bounds at fixed length and height)**.** *Let $S_0 = S_L = 0$, $S_t \ge 1$ for $0 < t < L$, $S_t - S_{t-1} \in \{-1, 0, 1\}$ and $H = \max S_t$. Then $L \ge 2H$ and*

$$H^2 + L - 2H \le \beta \le H(L-H), \tag{4.15}$$

$$\frac{2H^3+H}{3} + L - 2H \le Q \le \frac{2H^3+H}{3} + (L-2H)H^2. \tag{4.16}$$

All four bounds are attained. In particular

$$Q \ge \frac{2H^3+H}{3}, \tag{4.17}$$

and $Q \ge L - 1$.

Proof. Before the first visit to H and after it the path must visit each level $1, \dots, H-1$. Together with one visit to H, these $2H-1$ designated interior times contribute H^2 to β and $(2H^3+H)/3$ to Q. The remaining $L-2H$ times lie between heights 1 and H. Keeping them all at height 1 attains the lower bounds; keeping them at height H attains the upper bounds. □

Corollary 4.9 (Sharp fixed-length bounds)**.** *For $L \geq 2$,*

$$\beta \leq \left\lfloor \frac{L^2}{4} \right\rfloor,$$

and

$$Q \leq \begin{cases} \dfrac{L(L^2+2)}{12}, & L \text{ even}, \\ \dfrac{L(L^2-1)}{12}, & L \text{ odd}. \end{cases}$$

Both are attained by the tent path, with one zero step at the top when L is odd. In particular

$$Q \leq \frac{L^3+2L}{12}. \tag{4.18}$$

Proof. Use $S_t \leq \min(t, L-t)$ and sum this bound and its square. □

5. Where ζ enters

Status: MIXED. The residue formula for M is standard [18, §14.27], with the usual summation convention and multiplicity caveats. The displayed non-truncated expansion for $\widetilde{A}$ below is *FORMAL / NOT PROVED UNDER THE STATED HYPOTHESES*; it is retained only as a residue calculation. A rigorous use requires a truncated contour formula with an explicit remainder, or an additional hypothesis controlling negative moments of $\zeta'(\rho)$. Proposition 5.1 is an exact elementary identity proved here.

The black curve depends on M nonlinearly, through $\sqrt{H_\nu}$ and through the selection of u_ν, τ_ν, v_ν. No closed expression of its Mellin transform in terms of ζ is known to the author (version 5 asserted that none exists; that assertion is unproved and is withdrawn). The nodes u_ν, v_ν are zeros of M. If the nontrivial zeros are simple, the standard residue expansion (with no RH assumption) is

$$M(x) = \sum_\rho \frac{x^\rho}{\rho\,\zeta'(\rho)} - 2 + \sum_{n\geq 1} \frac{(-1)^{n-1}(2\pi/x)^{2n}}{(2n)!\,n\,\zeta(2n+1)}, \tag{5.1}$$

for non-integer x, the sum over ρ taken along a suitable sequence $|\gamma| < T_k \to \infty$ [18, §14.27]. For $\widetilde{A}(x) := \int_0^x M(t)\,dt$ one has, for $\operatorname{Re} s > 1$,

$$\int_1^\infty \widetilde{A}(x) x^{-s-2}\,dx = \frac{1}{s(s+1)\zeta(s)}. \tag{5.2}$$

The corresponding residue computation is

$$\widetilde{A}(x) \ \sim_{\text{formal}} \ \sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)\zeta'(\rho)} - 2x + 12 + (\text{residues from the trivial zeros}). \tag{5.3}$$

Status: FORMAL, NOT A THEOREM AS WRITTEN. No untruncated equality or $O(x^{-1})$ remainder is asserted here. The factor $(\rho+1)^{-1}$ improves convergence, but RH plus simplicity does not by itself control $1/\zeta'(\rho)$ strongly enough. Rigorous untruncated statements require additional input of Gonek–Hejhal type; see [16]. The area of M_{sk} differs from $\widetilde{A}$ by explicit local terms:

Proposition 5.1 (Dipole identity)**.** [unconditional] *For each excursion let its* legs *be the integer sets* $\lambda = (u_\nu, \tau_\nu] \cap \mathbb{Z}$ *and* $(\tau_\nu, v_\nu] \cap \mathbb{Z}$*, with midpoints* $c_\lambda = (u_\nu + \tau_\nu)/2$ *and* $(\tau_\nu + v_\nu)/2$*. Then for every zero* v_k*,*

$$\int_0^{v_k} M_{\mathrm{sk}}(t)\,dt = \widetilde{A}(v_k) + \sum_{\lambda \subset (0, v_k]} \sum_{m \in \lambda} \mu(m)\,(m - c_\lambda). \tag{5.4}$$

Proof. On an ascent leg, $\int_u^\tau M_{\mathrm{sk}} = \sigma H a/2$ and $\int_u^\tau M = \sum_{m=u+1}^{\tau} \mu(m)(\tau - m)$, while $\sigma H = \sum_{m=u+1}^{\tau} \mu(m)$; the difference is $\sum_m \mu(m)(a/2 - \tau + m) = \sum_m \mu(m)(m - c_\lambda)$. On a descent leg, $\int_\tau^v M_{\mathrm{sk}} = \sigma H b/2$ and $\int_\tau^v M = b\,\sigma H + \sum_{m=\tau+1}^{v} \mu(m)(v - m)$ with $\sum_{m=\tau+1}^{v} \mu(m) = -\sigma H$; the difference is $\sum_m \mu(m)(m - (v - b/2))$. Sum over legs. □

6. Bounds

Status: THEOREMS plus one withdrawn inference. The transfer argument is elementary; the input bounds for M are standard [19, 17, 14, 7]. The former inference from an excursion-height bound indexed by v_ν to a pointwise zero-free half-plane is *FALSE / WITHDRAWN*; see Remark 6.4.

Lemma 6.1 (Transfer)**.** [unconditional] *Let* $x \geq 1$ *and let* $x^* \geq x$ *be the first zero of* M *not smaller than* x *(or* N *in the terminal block). If* $|M(n)| \leq c\,n^{1/2}$ *for all* $1 \leq n \leq x^*$ *(so necessarily* $c \geq 1$*), then*

$$|M_{\mathrm{s}}(t)| \leq c^{1/2} t^{1/4} \quad (0 \leq t \leq x), \qquad \mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq \tfrac{4}{5}\, c^{1/2} x^{5/4}. \tag{6.1}$$

Proof. For $t \geq \tau_\nu$, $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq c^{1/2} \tau_\nu^{1/4} \leq c^{1/2} t^{1/4}$. For $u_\nu \leq t \leq \tau_\nu$ the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$ lies below the concave function $c^{1/2} t^{1/4}$, which is ≥ 0 at u_ν and $\geq \sqrt{H_\nu}$ at $\tau_\nu \leq x^*$. Integrate. □

Corollary 6.2. [unconditional] *From the Vinogradov–Korobov bound* [19]

$$M(x) \ll x \exp\!\big(-c(\log x)^{3/5} (\log\log x)^{-1/5}\big) \tag{6.2}$$

one obtains, for some $c' > 0$ *(not asserted to equal* $c/2$*),*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{3/2} \exp\!\big(-c'(\log x)^{3/5} (\log\log x)^{-1/5}\big). \tag{6.3}$$

Corollary 6.3. [unconditional implication] *If RH holds, then*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{5/4+\varepsilon}. \tag{6.4}$$

Consequently, if $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \gg x^{5/4+\varepsilon}$ *along a sequence, RH is false.*

Remark 6.4 (False inference withdrawn). **Status: FALSE / WITHDRAWN.** From $\ell_\nu \geq 2H_\nu$ and a bound $\sqrt{H_\nu}\ell_\nu \ll v_\nu^{5/4+\varepsilon}$ one may deduce only $H_\nu \ll v_\nu^{5/6+O(\varepsilon)}$. This is a bound indexed by the *future endpoint* v_ν; it does *not* imply $M(x) \ll x^{5/6+\varepsilon}$ at interior points, and therefore gives no zero-free half-plane. Version 5's claimed consequence $\operatorname{Re} s > 5/6$ was invalid. Whether (6.4) is equivalent to RH remains unresolved by the present argument.

Remark 6.5 (Oscillation of the heights). The strongest published numerical lower bounds recorded here are $\limsup M(x)/\sqrt{x} > 1.826054$ and $\liminf M(x)/\sqrt{x} < -1.837625$ [7]. In particular there are infinitely many n with $M(n) > 1.826054\sqrt{n}$; the excursion containing n has $u_\nu < n$, so $H_\nu > 1.826054\, u_\nu^{1/2}$ for infinitely many positive excursions, and similarly with 1.837625 for negative ones. Version 5 claimed $H_\nu \gg v_\nu^{1/2-\varepsilon}$; this is *not justified*, since v_ν may be much larger than n.

7. Criteria equivalent to RH

Status: MIXED. The Mellin identity and the global L^1/L^2 equivalences are classical consequences of Littlewood's criterion and partial summation; they are *not claimed as new RH criteria.* The excursion decompositions, pruning lemmas and local sufficient conditions are exact elementary repackagings proved here. Heuristics are explicitly unproved.

For $\operatorname{Re} s > 1$,

$$\int_1^\infty M(x)x^{-s-1}\,dx = \frac{1}{s\,\zeta(s)}. \tag{7.1}$$

If $\int_1^\infty |M(x)|x^{-\sigma-1}dx < \infty$ for every $\sigma > 1/2$, the left side continues analytically to $\operatorname{Re} s > 1/2$, so ζ has no zeros there.

Proposition 7.1 (L^1 criterion)**.** [unconditional] *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{v_\nu \le x} \beta_\nu + \beta_{\mathrm{term}}(x) = \sum_{n\le x} |M(n)| \ll x^{3/2+\varepsilon}. \tag{7.2}$$

Proof. RH gives $M(x) \ll x^{1/2+\varepsilon}$ [18, §14.25]. Conversely, by partial summation (7.2) gives $\sum_n |M(n)|n^{-\sigma-1} < \infty$ for $\sigma > 1/2+\varepsilon$, and M is constant on $[n, n+1)$; apply the remark after (7.1). □

Lemma 7.2 (Pruning)**.** [unconditional] *$\beta_\nu \le H_\nu \ell_\nu \le \ell_\nu^2/2$, hence*

$$\sum_{v_\nu \le x,\ \ell_\nu \le L_0} \beta_\nu \le \frac{xL_0}{2}. \tag{7.3}$$

Corollary 7.3 (Reduction to long excursions)**.** [unconditional] *For fixed $\delta \ge 0$ and $L_0 = x^{1/2-\delta}$, RH holds if and only if*

$$\sum_{v_\nu \le x,\ \ell_\nu > L_0} \beta_\nu + \beta_{\mathrm{term}}(x) \ll x^{3/2+\varepsilon} \quad \text{for every } \varepsilon > 0, \tag{7.4}$$

where the terminal block is included when its length exceeds L_0. At most x/L_0 excursions are involved.

Remark 7.4 (The $\sqrt{x}$ frontier)*.* The cut may even depend on ε: with $L_0 = x^{1/2+\varepsilon/2}$ the discarded part is $\le x^{3/2+\varepsilon/2}/2$. The frontier is therefore $x^{1/2+o(1)}$; what requires new information is a cut x^θ with a fixed $\theta > 1/2$ (version 5 stated the frontier as $x^{1/2}$).

Heuristic 7.5 (Renewal tail)**.** This is a heuristic argument. If excursion lengths behave like those of a recurrent random walk,

$$\#\{v_\nu \le x :\ \ell_\nu > L\} \asymp \sqrt{x}\,L^{-1/2}, \tag{7.5}$$

and with $\beta_\nu \asymp c_\beta \ell_\nu^{3/2}$ the dyadic range $[L, 2L]$ contributes $\asymp \sqrt{x}\,L$ to (7.2), so the terminal scales dominate.

Remark 7.6 (Limit of pruning)*.* For $\ell_\nu \asymp x$ the trivial bound gives $\beta_\nu \ll x^2$ against a budget $x^{3/2+\varepsilon}$: a factor $x^{1/2}$ is missing, and there are only $O(1)$ such excursions.

Theorem 7.7 (Correlation criterion)**.** [unconditional] *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{n\le x} M(n)^2 = \sum_{v_\nu \le x} Q_\nu + Q_{\mathrm{term}}(x) = -\sum_{v_\nu \le x}\sum_h hC_h^{(\nu)} + Q_{\mathrm{term}}(x) \ll x^{2+\varepsilon}. \tag{7.6}$$

Proof. RH gives $M \ll x^{1/2+\varepsilon}$. Conversely Cauchy–Schwarz gives $\sum_{n\le x}|M(n)| \le x^{1/2}(\sum M^2)^{1/2} \ll x^{3/2+\varepsilon/2}$; apply Proposition 7.1. □

Corollary 7.8 (Local sufficient condition, version 5)**.** [unconditional implication] *If for every $\varepsilon > 0$ there is C_ε with*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \textit{for every excursion (including the terminal block),} \tag{7.7}$$

then RH holds.

Heuristic 7.9 (The hypothesis of Corollary 7.8 fails)**.** This is a heuristic argument. In a model where, after a zero of M, the next values of μ behave locally at random with squarefree density $6/\pi^2$, the pattern "k non-zero steps of one sign followed by k of the other" follows a given zero with probability $\ge c_0^{-k}$. Even with infinitely many zeros, this model does *not* by itself prove that such patterns occur for unbounded k; that would require a quantitative number of trials together with dependence control. The statement that excursions with $H_\nu = k$ and $\ell_\nu \asymp k$ should occur for $k \to \infty$ is therefore only a Borel–Cantelli-style heuristic prediction. By (4.17) $Q_\nu/\ell_\nu^{2+\varepsilon} \gg k^{1-\varepsilon} \to \infty$, so (7.7) fails for every $\varepsilon \in (0,1)$; for $\varepsilon \ge 1$ it is trivially true by (4.18). Version 5 called the condition "sufficient but not weak"; heuristically it is never satisfied.

Corollary 7.10 (Pruned local condition)**.** [unconditional implication] *If for every $\varepsilon > 0$ there is C_ε with*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \textit{whenever } \ell_\nu > v_\nu^{1/2} \tag{7.8}$$

(and likewise for the terminal block when its length exceeds $x^{1/2}$), then RH holds.

Proof. By (4.18), excursions with $\ell_\nu \le x^{1/2}$ satisfy $Q_\nu \le (\ell_\nu^3 + 2\ell_\nu)/12 \le \ell_\nu x/12 + \ell_\nu/6$. The excursion interiors are disjoint, so their total length is at most x; the short excursions therefore contribute at most $x^2/12 + x/6$. The others contribute $\le C_\varepsilon \sum \ell_\nu^{2+\varepsilon} \le C_\varepsilon x^{2+\varepsilon}$. Apply Theorem 7.7. □

The ballistic excursions of Heuristic 7.9 have $\ell_\nu \asymp k \ll v_\nu^{1/2}$ and do not contradict (7.8).

Remark 7.11 (Scale of the correlation sums)*.* Under the crude model in which "j squarefree" and "$j+h$ squarefree" are independent with probability $6/\pi^2$ each, $P_h \approx (6/\pi^2)^2(L-h)$ and

$$\sum_{h<L} hP_h \approx \Big(\frac{6}{\pi^2}\Big)^2 \sum_{h<L} h(L-h) \approx \frac{6}{\pi^4} L^3 \approx 0.0616\, L^3. \tag{7.9}$$

Version 5 gave $\frac{1}{2}(6/\pi^2)^2 L^3 \approx 0.18 L^3$, omitting the factor $L-h$; that constant is *incorrect*, the order L^3 stands. For $|\sum_h hC_h|$ of size $B(\sum_h h^2)^{1/2} \asymp BL^{3/2}$ to be $O(L^2)$ one needs $B = O(\sqrt{L})$ (a calibration, not a proof).

Remark 7.12 (Conditioning on the bridge)*.* Inside an excursion the correlations are constrained by (4.6), (4.7) and by the geometry of the bridge; with the defect $d_\nu := \ell_\nu/2 - H_\nu \ge 0$, a third constraint of this type is a research direction, not a result.

Theorem 7.13 (Energy criterion)**.** [unconditional] *Define, with u the last zero $\le x$ and $H_{\mathrm{run}}(x) = \max_{u<n\le x} |M(n)|$,*

$$\mathcal{E}(x) := \frac{1}{4}\Big(\sum_{v_\nu \le x} H_\nu \ell_\nu + (x-u) H_{\mathrm{run}}(x) \Big). \tag{7.10}$$

For a complete excursion $\frac{1}{4} H_\nu \ell_\nu = \int_{I_\nu} (J')^2$, where J interpolates $\mathcal{A}$ linearly between consecutive zeros; for the terminal block this interpretation does not hold and (7.10) *is a definition. Then*

$$\sum_{n\le x} |M(n)| \le 4\mathcal{E}(x) \le x \max_{n\le x} |M(n)|, \tag{7.11}$$

and RH holds if and only if $\mathcal{E}(x) \ll x^{3/2+\varepsilon}$ for every $\varepsilon > 0$.

Proof. $\beta_\nu \le H_\nu \ell_\nu$ and $\sum \ell_\nu \le x$; apply Proposition 7.1. □

8. Lacunary long excursions and localized RH criteria

Status: THEOREM. Lemmas 8.2 and 8.3 are unconditional geometric counting bounds, using only disjointness and ordering of excursions. The arithmetic input is Corollary 4 of Kaczorowski–Pintz [9]: at $a = 0$ their function is $M_0 = M + 2$, not M. The proof below explicitly accounts for this additive constant. Subject to that published theorem, the localized RH criteria below are unconditional theorems.

Put

$$\Theta = \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}, \qquad \gamma_1 = 14.134725\ldots.$$

Definition 8.1 (Power- and linear-long excursions)**.** For $A > 1$, $I_\nu = (x_{0,\nu}, x_{1,\nu}]$ is *A-exceptional* if $x_{1,\nu} > x_{0,\nu}^A$ and A-ordinary otherwise. For $\lambda > 1$ it is *λ-long* if $x_{1,\nu} > \lambda x_{0,\nu}$. The initial excursion $(0, 2]$ is treated as one finite exceptional case.

Lemma 8.2 (Power lacunarity: the $O(\log\log X)$ bound)**.** *For every $A > 1$,*

$$E_A(X) := \#\{\nu : x_{0,\nu} \le X,\ x_{1,\nu} > x_{0,\nu}^A\} \le 2 + \frac{1}{\log A}\log\frac{\log X}{\log 2}. \tag{8.1}$$

Thus $E_A(X) = O_A(\log\log X)$. In particular, for $A = 2$,

$$E_2^{\mathrm{pow}}(X) \le 2 + \log_2\frac{\log X}{\log 2}. \tag{8.2}$$

Proof. List exceptional starts at least 2 as $2 \le a_1 < a_2 < \cdots$. Since the next excursion starts at or after the end of the preceding one, possibly after a zero run, $a_{j+1} > a_j^A$. Hence $\log a_j > A^{j-1}\log 2$. □

Lemma 8.3 (Linear lacunarity: the $O(\log X)$ bound)**.** *For every $\lambda > 1$ and $X \ge 2$,*

$$E_\lambda^{\mathrm{lin}}(X) := \#\{\nu : x_{0,\nu} \le X,\ x_{1,\nu} > \lambda x_{0,\nu}\} \le 2 + \frac{\log(X/2)}{\log\lambda}. \tag{8.3}$$

In particular the excursions with $x_1 > 2x_0$ are at most $(\log 2)^{-1}\log X + O(1)$.

Proof. For starts $a_j \ge 2$, disjointness gives $a_{j+1} > \lambda a_j$, hence $a_j > 2\lambda^{j-1}$. □

Theorem 8.4 (Kaczorowski–Pintz, specialized to M)**.** ***Status: THEOREM (published input).*** *Let $M_0(x) = M(x) + 2$. Corollary 4 of* [9] *gives*

$$\liminf_{Y\to\infty}\frac{V(M_0, Y)}{\log Y} \ge \frac{\gamma_1}{\pi}, \tag{8.4}$$

with combined oscillation of size $x^{\Theta-\eta}$ for every $\eta > 0$. Consequently, if $\Theta > 1/2$ and $0 < \eta < \Theta - 1/2$, then

$$N_\eta(Y) \ge \left(\frac{\gamma_1}{\pi} - o(1)\right)\log Y, \tag{8.5}$$

where $N_\eta(Y)$ counts distinct excursions starting at most Y that contain a point $t \le Y$ with

$$|M(t)| \ge \tfrac{1}{2}t^{\Theta-\eta}.$$

Proof. At $a = 0$, the function M_a of [9] is $M_0 = M - 1/\zeta(0) = M + 2$. The combined oscillation points alternate sign and have $|M_0(t)| \ge t^{\Theta-\eta}$. For sufficiently large t the positive points have $M(t) > 0$ and the negative points have $M(t) < 0$; therefore consecutive selected points, and hence all selected alternating points, lie in distinct excursions. Their starts are below the points and thus at most Y. The additive constant 2 is absorbed by the factor $1/2$. □

Theorem 8.5 (Sparse-exception localization of RH)**.** ***Status: THEOREM.*** *Let $\mathcal{E}$ be any family of excursions satisfying*

$$\#\{\nu \in \mathcal{E} : x_{0,\nu} \le X\} = o(\log X).$$

Then RH is equivalent to: for every $\varepsilon > 0$ there exists C_ε such that

$$|M(t)| \le C_\varepsilon t^{1/2+\varepsilon} \tag{8.6}$$

for every t lying in every excursion outside $\mathcal{E}$. No bound of any kind is required on the excursions in $\mathcal{E}$.

Proof. RH implies the global estimate by Littlewood. Conversely, suppose $\Theta > 1/2$ and choose $\eta, \varepsilon > 0$ with $1/2 + \varepsilon < \Theta - \eta$. Theorem 8.4 supplies $\gg \log X$ distinct excursions containing values $\gg t^{\Theta-\eta}$, whereas only $o(\log X)$ may belong to $\mathcal{E}$. Arbitrarily large points outside $\mathcal{E}$ therefore violate (8.6). □

Corollary 8.6 (Power-localized RH criterion)**.** ***Status: THEOREM.*** *For every fixed $A > 1$, RH is equivalent to* (8.6) *restricted only to A-ordinary excursions $x_1 \le x_0^A$. In particular,*

$$\boxed{\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \text{ for all } t \text{ in excursions with } x_1 \le x_0^2.} \tag{8.7}$$

No estimate at all is required on the excursions with $x_1 > x_0^2$.

Proof. Lemma 8.2 gives $E_A(X) = O_A(\log\log X) = o(\log X)$. Apply Theorem 8.5. □

Remark 8.7 (What the $O(\log\log X)$ theorem does and does not bound). **Status: NO COTA on individual height from lacunarity alone.** The theorem $E_A(X) = O_A(\log\log X)$ bounds only the *number* of A-exceptional excursions. It gives no upper or lower RH-scale bound for H_ν, β_ν or Q_ν of an individual exceptional excursion. No probabilistic tail estimate is used anywhere in Corollary 8.6. In particular the previously mentioned $\sqrt{\log\log\log X}$ extreme-value scale is a heuristic from a random model, not a theorem, and is not used in this manuscript.

Corollary 8.8 (Linear-localized RH criterion)**.** ***Status: THEOREM.*** *Let*

$$\lambda_* := \exp(\pi/\gamma_1) = 1.2488968\ldots.$$

For every fixed $\lambda > \lambda_*$, RH is equivalent to (8.6) *restricted only to excursions with $x_1 \le \lambda x_0$. In particular it is enough to control excursions with $x_1 \le 2x_0$.*

Proof. The complementary family has at most $(1/\log\lambda + o(1))\log X$ members by Lemma 8.3. If RH fails, Theorem 8.4 gives at least $(\gamma_1/\pi - o(1))\log X$ large distinct excursions. The strict inequality $1/\log\lambda < \gamma_1/\pi$ leaves a positive proportion outside the long family. □

Remark 8.9 (Role of the two new counts). The $O(\log\log X)$ and $O(\log X)$ bounds are geometric, not probabilistic. Their arithmetic force appears after combination with the independent Kaczorowski–Pintz lower bound of order $\log X$ for large alternating oscillations. The exponent 2 is not needed for localization itself; it is singled out next because the range $t \le x_0^2$ has an exact bilinear kernel.

8.1. **The square range as an exact bilinear form of the prefix.**

Proposition 8.10 (Bilinear square-range identity)**.** *Let $u \ge 1$ and $1 \le t \le u^2$. Then*

$$M(t) = 2M(\min(t,u)) - \sum_{d,j \le u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor. \tag{8.8}$$

In particular, if $M(u) = 0$ and $u < t \le u^2$,

$$M(t) = -B_u(t), \qquad B_u(t) := \sum_{d,j \le u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor, \qquad B_u(u) = 0. \tag{8.9}$$

Proof. Let $\mu_u = \mu \mathbf{1}_{[1,u]}$. The convolution $(\mu-\mu_u)*(\mu-\mu_u)*1$ is supported on (u^2,∞). Expanding and using $\mu*1=\delta$ gives $\mu(n)=2\mu_u(n)-(\mu_u*\mu_u*1)(n)$ for $n\le u^2$; summing gives the formula. □

Remark 8.11 (Higher-order determinacy; why the square is special). For every $k\ge 1$ and $n\le u^k$,

$$\mu(n)=\sum_{j=1}^{k}(-1)^{j-1}\binom{k}{j}(\mu_u^{*j}*1^{*(j-1)})(n). \tag{8.10}$$

Thus "determined by the prefix" is not special to u^2. What is special to $t\le u^2$ is the order-2 form (8.9), with rank-one coefficients $\mu(d)\mu(j)$ and elementary kernel $\lfloor t/(dj)\rfloor$. This is the $k=2$ Heath–Brown/Vaughan-type identity; related summed forms are used in fast computation of M [3, 4].

Question 8.12 (Central square-excursion problem)**. Status: OPEN; NO RH-SCALE BOUND IS PROVED HERE.** Let $u=x_{0,\nu}$ and $x_{1,\nu}\le u^2$. Then $B_u(u)=B_u(x_{1,\nu})=0$ and $-\sigma_\nu B_u(t)>0$ for $u<t<x_{1,\nu}$. Can the rank-one coefficient matrix $\mu(d)\mu(j)$, the product kernel $\lfloor t/(dj)\rfloor$, and these bridge/sign constraints be combined to prove

$$\max_{u<t\le x_{1,\nu}}|B_u(t)|\ll_\varepsilon x_{1,\nu}^{1/2+\varepsilon}$$

uniformly over all such excursions? By Corollary 8.6, this is equivalent to RH. The point of the formulation is that it isolates the entire problem inside finite bilinear bridges of the square range.

8.2. **Mellin mass of a complete excursion: the direction $M\to\zeta$.**

Proposition 8.13 (Exact excursion block and its Mellin mass)**.** *For a complete excursion $I=(x_0,x_1]$, put*

$$D_I(s)=\sum_{x_0<n\le x_1}\frac{\mu(n)}{n^s}.$$

Then

$$D_I(s)=\sum_{n=x_0+1}^{x_1-1}M(n)\big(n^{-s}-(n+1)^{-s}\big). \tag{8.11}$$

For real $\sigma>0$, $D_I(\sigma)$ has the sign of the excursion and

$$|D_I(\sigma)|\ge\sigma\beta_I x_1^{-\sigma-1}\ge\sigma H_I^2 x_1^{-\sigma-1}. \tag{8.12}$$

For $\Re s>1$,

$$\frac{1}{\zeta(s)}=\sum_I D_I(s), \tag{8.13}$$

zero runs contributing nothing.

Proof. Summation by parts gives (8.11) because the endpoint values of M vanish. On an excursion all interior $M(n)$ have the same sign, and $n^{-\sigma}-(n+1)^{-\sigma}\ge\sigma x_1^{-\sigma-1}$. The bound $\beta_I\ge H_I^2$ follows from Proposition 4.8. Summing the disjoint Dirichlet blocks gives (8.13) in the half-plane of absolute convergence. □

Remark 8.14 (No finite-data assumption). Nothing in this section treats the computed range of M as representative. The statements are uniform in x_0; a dangerous excursion may begin at 10^{40} or arbitrarily farther out. The role of x_0^2 is algebraic, not empirical.

9. Trial-division horizon of an excursion

Status. Lemma 9.1 is standard trial division. The $O(\log\log X)$ separation has already been isolated in Lemma 8.2; Proposition 9.6 records its $A = 2$ form in the historical notation of versions 5–6.

Lemma 9.1 (Square horizon)**.** [unconditional] *Let $u \geq 2$. Trial division by the primes $p \leq u$ alone determines $\mu(n)$ for every $n \leq u^2$.*

Proof. Remove the prime factors $p \leq u$ of n; if one occurs twice, $\mu(n) = 0$. The cofactor $q \leq n \leq u^2$ has all prime factors $> u$, so it is 1 or a single prime (two such factors would give $q > u^2$). □

Warning 9.2 (On the word "information")*.* Primes $> u$ are not logically new information, and prefix determinacy is not special to u^2: the higher-order identity (8.10) propagates a finite prefix to every fixed power range. The square is special only in the more concrete sense that one-stage trial division by primes $\leq u$ suffices and the exact summatory formula has the bilinear kernel $\lfloor t/(dj)\rfloor$. Historical uses of "known", "new" and "information" below refer only to this finite-level decomposition, not to logical independence.

Definition 9.3 (Smooth and rough parts)**.** For $y \geq 1$ write $n = s_y(n)\, r_y(n)$, where every prime factor of $s_y(n)$ is $\leq y$ and every prime factor of $r_y(n)$ is $> y$. Let $P^+(n)$ be the largest prime factor of n, with $P^+(1) = 1$.

Lemma 9.4. [unconditional]

$$\mu(n) = \mu(s_y(n))\,\mu(r_y(n)); \tag{9.1}$$

if $n < y^k$ then $\Omega(r_y(n)) \leq k-1$, and if $n \leq y^2$ then $r_y(n)$ is 1 or a prime.

Proof. The two parts are coprime. If $\Omega(r) \geq 1$ then $y^{\Omega(r)} < r \leq n$. □

Definition 9.5 (Square-horizon classes)**.** An excursion is *square-ordinary* if $v_\nu \leq u_\nu^2$ and *square-exceptional* otherwise. These replace the historical information terminology of versions 5–6.

For $u_\nu \geq 2$,

$$\vartheta_\nu = \min\left\{1, \frac{u_\nu^2 - u_\nu}{\ell_\nu}\right\}$$

is the fraction of the excursion lying below the square horizon.

Proposition 9.6 (Double-exponential separation)**.** *Let $a_1 < a_2 < \cdots$ be the starting points at least 2 of square-exceptional excursions. Then*

$$a_{j+1} > a_j^2, \qquad a_j \geq a_1^{2^{j-1}},$$

and

$$N_{\mathrm{sq}}(X) = \#\{j : a_j \leq X\} \leq 1 + \log_2 \frac{\log X}{\log 2}.$$

This is the $A = 2$ case of Lemma 8.2.

Proof. The end of a square-exceptional excursion starting at a_j is beyond a_j^2, and the next such excursion starts at or after that end. □

Question 9.7. Status: OPEN IN THE PRESENT ARGUMENT; NO FINITENESS THEOREM IS CLAIMED. Are there only finitely many square-exceptional excursions? Equivalently: does M vanish in $(x, x^2]$ for every sufficiently large x? (If the zeros $u \geq x_0$ all have a zero in $(u, u^2]$, apply this to the last zero $\leq x$; the converse is immediate.) Kaczorowski–Pintz Corollary 4 does *not* settle this question at $a = 0$: its sign-change function is $M_0 = M + 2$, so a sign change of M_0 does not force a zero of M. No finiteness conclusion is drawn here. This sentence is a statement about the present proof, not a claim of exhaustive bibliographic priority.

Remark 9.8 (The horizon u^2 is not the multiplicative threshold of an excursion). For a square-ordinary excursion $\sqrt{v_\nu} \le u_\nu$. Every $n \le v_\nu$ has at most one prime factor $> \sqrt{v_\nu}$, so the threshold that governs the multiplicative structure inside I_ν is $\sqrt{v_\nu}$, which lies below u_ν. The split "known/new" at u_ν does not match it, and for a square-ordinary excursion the regime $t > u_\nu^2$ of Section 10 lies beyond v_ν. That section is therefore about $M(t)$, not about a single excursion; Section 11 gives the local version.

10. Large-prime decomposition at level u^2

Status: THEOREMS, with classical sieve ancestry. The identities are elementary decompositions by large prime factors and are instances of Buchstab/Legendre-type sieve organization; no blanket claim of originality is made. The asymptotic statements use only the prime number theorem with classical error term and the Vinogradov–Korobov bound [15, 19], and are proved here. Several statements of version 5 are corrected; see Warnings 10.9, 10.13, 10.18 and 10.19.

Definition 10.1 (Baseline and large-prime part). For $u \ge 2$ and $t \ge 1$ put

$$B_u(t) := \sum_{\substack{n \le t \\ P^+(n) \le u^2}} \mu(n), \qquad E_u(t) := \sum_{\substack{n \le t \\ P^+(n) > u^2}} \mu(n), \qquad M(t) = B_u(t) + E_u(t). \tag{10.1}$$

Version 5 used an undefined "baseline" $B_u(t)$; (10.1) is the definition adopted here. In the window $u^2 < t < u^4$, E_u is the quantity $E_u^{(1)}$ of version 5.

Proposition 10.2 (Single-prime copy). [unconditional] *For a prime p and $t < p^2$,*

$$\Delta_p(t) := \sum_{a \le t/p} \mu(ap) = -M(t/p). \tag{10.2}$$

Only $p > \sqrt{t}$ is used; the condition $p > u^2$ of version 5 plays no role.

Proof. $a \le t/p < p$, so $p \nmid a$ and $\mu(ap) = -\mu(a)$. □

Proposition 10.3 (Sign content of one prime cone). [unconditional] *Let $\mathcal{S}(y) = \sum_{a \le y} \mu(a)^2$. Among ap, $a \le y < p$, the numbers of values $+1, -1, 0$ of μ are*

$$N_+ = \frac{\mathcal{S}(y) - M(y)}{2}, \qquad N_- = \frac{\mathcal{S}(y) + M(y)}{2}, \qquad N_0 = \lfloor y \rfloor - \mathcal{S}(y), \tag{10.3}$$

with limiting proportions $3/\pi^2,\ 3/\pi^2,\ 1 - 6/\pi^2$.

Proposition 10.4 (Linear window). [unconditional] *If $u^2 < t < u^4$, no $n \le t$ has two prime factors $> u^2$ (counted with multiplicity), and*

$$E_u(t) = -\sum_{u^2 < p \le t} M(t/p). \tag{10.4}$$

Proposition 10.5 (Prime-count form). [unconditional] *For $u^2 < t < u^4$ and $R := t/u^2$,*

$$E_u(t) = -\sum_{a \le R} \mu(a)(\pi(t/a) - \pi(u^2)), \qquad M(t) = B_u(t) - \sum_{a \le R} \mu(a)\pi(t/a) + M(R)\pi(u^2). \tag{10.5}$$

Proof. Expand (10.4) as $-\sum_p \sum_{a \le t/p} \mu(a)$ and interchange: for fixed a the prime runs over $u^2 < p \le t/a$, which is non-empty only if $a < R$. □

Remark 10.6 (How the primes above u^2 enter). The only input from primes $> u^2$ is the value of π at the points t/a, $a \le R$, weighted by $\mu(a)$ at the small scale $R < u^2$. Individual positions of those primes matter only through these counts.

Definition 10.7 (Copy multiplicities)**.** For $u^2 < t < u^4$, $R = t/u^2$ and $1 \le y \le \lfloor R \rfloor$ put

$$N_y(t,u) = \#\Big\{p:\ \max(u^2, \tfrac{t}{y+1}) < p \le \tfrac{t}{y}\Big\}, \qquad E_u(t) = -\sum_{1\le y\le R} N_y(t,u) M(y). \tag{10.6}$$

Remark 10.8 (Asymptotics of the multiplicities, corrected). For fixed $R > 1$ and $u \to \infty$,

$$N_y \sim \frac{Ru^2}{2\log u}\cdot\frac{1}{y(y+1)} \quad (1 \le y \le R-1), \qquad N_{\lfloor R\rfloor} \sim \frac{u^2}{2\log u}\Big(\frac{R}{\lfloor R\rfloor} - 1\Big) \quad (R \notin \mathbb{Z}), \tag{10.7}$$

and $N_R = 0$ if $R \in \mathbb{Z}$.

Warning 10.9. Version 5 stated the first relation of (10.7) for all $y < R$. This is *incorrect* for $y = \lfloor R \rfloor$ when $R \notin \mathbb{Z}$, since then $t/(y+1) < u^2$.

Definition 10.10 (Coherent coefficient)**.**

$$\mathcal{C}(R) := \int_1^R \frac{M(w)}{w^2}\,dw = \sum_{a\le R}\mu(a)\Big(\frac{1}{a} - \frac{1}{R}\Big) = \sum_{a\le R}\frac{\mu(a)}{a} - \frac{M(R)}{R}. \tag{10.8}$$

Proposition 10.11 (Fixed R, corrected)**.** [unconditional] *For fixed $R > 1$, as $u \to \infty$,*

$$E_u(Ru^2) = -\frac{Ru^2}{2\log u}\big(\mathcal{C}(R) + o(1)\big). \tag{10.9}$$

Proof. In (10.5) the sum is finite; for $a < R$ the prime number theorem gives $\pi(Ru^2/a) - \pi(u^2) \sim (R/a - 1)u^2/(2\log u)$, and $\sum_{a\le R}\mu(a)(R/a-1) = R\,\mathcal{C}(R)$. □

Lemma 10.12 (The coefficient of version 5)**.** [unconditional] *The coefficient $A(R)$ of version 5, here*

$$\Theta(R) := \sum_{1\le y\le R}\frac{M(y)}{y(y+1)} = \sum_{n\le R}\frac{\mu(n)}{n} - \frac{M(R)}{\lfloor R\rfloor + 1}, \tag{10.10}$$

satisfies

$$\Theta(R) = \mathcal{C}\big(\lfloor R\rfloor + 1\big), \qquad \mathcal{C}(R) = \Theta(R) - M(R)\Big(\frac{1}{R} - \frac{1}{\lfloor R\rfloor + 1}\Big), \tag{10.11}$$

and

$$\mathcal{C}(R) = -\int_R^\infty \frac{M(w)}{w^2}\,dw, \qquad \Theta(R) = -\sum_{n>R}\frac{M(n)}{n(n+1)}. \tag{10.12}$$

Version 5 wrote $M(R)/(R+1)$ in (10.10)*; this is correct only for integer R.*

Proof. M is constant on $[y, y+1)$ and $\int_y^{y+1} w^{-2}dw = 1/(y(y+1))$. By (7.1) and dominated convergence as $s \downarrow 1$ (using $M(w) \ll we^{-c\sqrt{\log w}}$), $\int_1^\infty M(w)w^{-2}dw = 0$. □

Warning 10.13 (Main term of version 5). Version 5 asserted $E_u^{(1)}(Ru^2) \sim -\frac{Ru^2}{2\log u}\Theta(R)$. By (10.9) and (10.11) this is *correct only when* $M(\lfloor R\rfloor) = 0$. For $1 < R < 2$ one has $E_u = -(\pi(t) - \pi(u^2)) \sim -(R-1)\frac{u^2}{2\log u}$, whereas version 5 gives $-\frac{R}{2}\cdot\frac{u^2}{2\log u}$.

Corollary 10.14 (Decay of the coefficients)**.** [unconditional] *By* (10.12) *and* (6.2)*,*

$$\mathcal{C}(R),\ \Theta(R) \ll \exp\big(-c_1(\log R)^{3/5}(\log\log R)^{-1/5}\big). \tag{10.13}$$

Proposition 10.15 (Both parts are large)**.** [unconditional]

(i) *For fixed $R > 1$ with $\mathcal{C}(R) \neq 0$,*

$$E_u(Ru^2) \sim -\frac{Ru^2}{2\log u}\mathcal{C}(R), \qquad B_u(Ru^2) \sim +\frac{Ru^2}{2\log u}\mathcal{C}(R); \tag{10.14}$$

for $1 < R < 2$, $\mathcal{C}(R) = 1 - 1/R > 0$.

(ii) *There is $C > 0$ such that, as $t \to \infty$, uniformly for $u^2 < t < u^4$ with $R = t/u^2 \geq \exp(C(\log\log t)^2)$,*

$$E_u(t) = (1 + o(1))\frac{t}{(\log t)^2}, \qquad B_u(t) = -(1 + o(1))\frac{t}{(\log t)^2}. \tag{10.15}$$

In both regimes $|E_u(t)|$ and $|B_u(t)|$ exceed $t^{1/2+\varepsilon}$ for every fixed $\varepsilon < 1/2$ and all large t.

Proof. We use [15]: for some $c > 0$,

$$\pi(y) = \mathrm{li}(y) + O\big(ye^{-c\sqrt{\log y}}\big), \quad M(y) \ll ye^{-c\sqrt{\log y}}, \quad \widetilde{m}(y) := \sum_{a\leq y}\frac{\mu(a)}{a} \ll e^{-c\sqrt{\log y}}, \tag{10.16}$$

and $\sum_{n\geq 1}\mu(n)\log n/n = -1$ (the value at $s = 1$ of $\zeta'(s)/\zeta(s)^2$). Hence

$$\int_1^\infty \widetilde{m}(w)\frac{dw}{w} = \lim_{Y\to\infty}\Big(\widetilde{m}(Y)\log Y - \sum_{a\leq Y}\frac{\mu(a)\log a}{a}\Big) = 1, \tag{10.17}$$

with absolute convergence.

(i) Proposition 10.11, and $M(t) = o(t/\log t)$. For $1 < R < 2$, $M = 1$ on $[1, 2)$.

(ii) Since $t < u^4$, $R < \sqrt{t}$, so $\log(t/a) \geq \frac{1}{2}\log t$ for $a \leq R$. By (10.5),

$$E_u(t) = -\sum_{a\leq R}\mu(a)\,\mathrm{li}(t/a) - \sum_{a\leq R}\mu(a)(\pi(t/a) - \mathrm{li}(t/a)) + M(R)\pi(u^2). \tag{10.18}$$

The second term is $\ll t(1 + \log R)e^{-c\sqrt{\log t/2}} = o(t/\log^2 t)$. The third is $\ll Re^{-c\sqrt{\log R}}\cdot u^2/\log u \ll e^{-c\sqrt{\log R}}\,t/\log t$, which is $o(t/\log^2 t)$ once $c\sqrt{C} > 1$. For the first, with $F(w) = w\,\mathrm{li}(t/w)$,

$$\sum_{a\leq R}\mu(a)\,\mathrm{li}(t/a) = \widetilde{m}(R)F(R) - \int_1^R \widetilde{m}(w)F'(w)\,dw, \tag{10.19}$$

$$\begin{aligned} F'(w) &= \mathrm{li}(t/w) - \frac{t/w}{\log(t/w)} \\ &= \frac{t/w}{\log^2(t/w)}\Big(1 + O\Big(\frac{1}{\log t}\Big)\Big). \end{aligned}$$

The boundary term is $\ll e^{-c\sqrt{\log R}}t/\log t = o(t/\log^2 t)$. The integral equals $\frac{t}{\log^2 t}\big(\int_1^R \widetilde{m}(w)g_t(w)\frac{dw}{w} + O(1/\log t)\big)$ with $g_t(w) = \log^2 t/\log^2(t/w) \in [1, 4]$ and $g_t \to 1$ pointwise; by dominated convergence and (10.17) it is $(1 + o(1))t/\log^2 t$. Hence $E_u(t) = (1 + o(1))t/\log^2 t$, and $B_u = M - E_u$ with $M(t) = o(t/\log^2 t)$. □

Remark 10.16 (Why $t/\log^2 t$). $\mathcal{C}(R)$ captures only the $1/\log t$ term, which tends to zero. The $1/\log^2 t$ term comes from $\zeta'(s)/\zeta(s)^2 \to -1$, that is, from the pole of ζ at $s = 1$, and it does not decay in R. The version 5 claim that the "coherent fraction tends to zero" is true for the first term and does not describe the size of E_u.

Definition 10.17 (Remaining margin, version 5)**.** For a target $T(t) = t^{1/2+\varepsilon}$ put

$$G_{u,\varepsilon}(t) = \max\big(0,\ T(t) - |B_u(t)|\big), \tag{10.20}$$

and call the large-prime part *causally responsible* for a crossing of T if $|E_u(t)| \geq G_{u,\varepsilon}(t)$ at the crossing.

Warning 10.18 (The margin is vacuous). With B_u as in (10.1), Proposition 10.15 gives $|B_u(t)| > T(t)$, hence $G_{u,\varepsilon}(t) = 0$, for all large t in both regimes. The test is always passed and does not discriminate. "Causal responsibility" of one of two cancelling terms of size $t/\log^k t$ for the value of their sum is not well defined.

Warning 10.19 (Calibration remarks of version 5). Version 5 required $|A(R)| \ll t^{-1/2+\varepsilon}\log t$ for the large-prime part to stay at the RH scale, and for $t = u^\alpha$ derived $|A(R)| \ll R^{-\eta(\alpha,\varepsilon)}$ with $\eta = \alpha(1/2-\varepsilon)/(\alpha-2)$ ($\eta = 3/2, 1, 3/4, 5/8$ at $\alpha = 3, 4, 6, 10$). These remarks are *superseded*: the coefficient is $\mathcal{C}$, not Θ (Warning 10.13); for $\alpha > 4$ the point t lies outside the window of Proposition 10.4, where (10.2) fails for $u^2 < p \le \sqrt{t}$; and by Proposition 10.15, E_u alone is never at the RH scale, so only $B_u + E_u = M$ can be compared with T.

Remark 10.20 (First copy transitions). For $1 < R < 2$ every copy has length 1 and $E_u(t) = -(\pi(t) - \pi(u^2))$, consistent with $\mathcal{C}(R) = 1 - 1/R$. For $2 < R < 3$ copies of length 2 appear but $M(2) = 0$ ($\mathcal{C} = 1/2$). For $3 < R < 4$ copies of length 3 enter with $M(3) = -1$, of opposite net sign.

11. Local large-prime decomposition of an excursion

Status: THEOREMS. Proposition 11.1 and Lemma 11.3 are proved here; the decomposition has classical sieve/Buchstab ancestry. The input on primes in short intervals is from the literature [13]; Heuristic 11.4 is heuristic.

Proposition 11.1 (Local decomposition)**.** [unconditional] *Let $I_\nu = (u, v]$ be a complete excursion and $y = \sqrt{v}$. For $u < t \le v$,*

$$M(t) = \sum_{\substack{u<n\le t\\ P^+(n)\le y}} \mu(n) - \sum_{a<y} \mu(a)\,\Pi_a(t), \qquad \Pi_a(t) := \#\Big\{p:\ \max\big(y, \tfrac{u}{a}\big) < p \le \tfrac{t}{a}\Big\}, \tag{11.1}$$

and in particular the bridge condition $M(v) = 0$ reads

$$\sum_{\substack{u<n\le v\\ P^+(n)\le \sqrt{v}}} \mu(n) = \sum_{a<\sqrt{v}} \mu(a)\,\Pi_a(v). \tag{11.2}$$

Proof. $M(u) = 0$. An $n \le v = y^2$ has at most one prime factor $p > y$; then $n = ap$ with $a < v/y = y < p$, so $\mu(n) = -\mu(a)$, and $(a, p) \mapsto ap$ is a bijection onto these n. □

Remark 11.2 (What the large primes do inside an excursion). The decomposition holds for every excursion, with threshold $\sqrt{v_\nu}$ and no horizon u_ν^2. The large-prime input consists of prime counts in the intervals $(u/a, t/a]$, of length at most ℓ_ν/a at height about v_ν/a. Historically Huxley proved the prime number theorem uniformly in all intervals of length $h \ge x^{7/12+\varepsilon}$ [13]. This is no longer the record: the zero-density estimates of Guth–Maynard imply the prime number theorem in all intervals of length $x^{17/30+o(1)}$ [5]. For Möbius sums, Matomäki–Teräväinen prove $\sum_{x<n\le x+x^\theta} \mu(n) = o(x^\theta)$ for every $\theta > 0.55$ [6]. These are the current inputs recorded here; below those ranges this paragraph asserts *NO COTA* of PNT-asymptotic strength.

Lemma 11.3 (Inverse logarithms)**.** [unconditional] *With C as in Proposition 10.15, the following holds uniformly for*

$$\exp(C(\log\log x)^2) \le z \le \sqrt{x}:$$

$$\sum_{a\le z} \frac{\mu(a)}{a\log(x/a)} = -\frac{1+o(1)}{(\log x)^2}. \tag{11.3}$$

Proof. Partial summation gives $\widetilde{m}(z)/\log(x/z) - \int_1^z \widetilde{m}(w)\,dw/(w\log^2(x/w))$; argue as in the proof of Proposition 10.15. □

Heuristic 11.4 (Opposite drifts)**.** This is a heuristic argument. Let $u \geq \sqrt{v}\exp(C(\log\log v)^2)$. Near $t \in I_\nu$ the counts Π_a grow for $a < t/\sqrt{v}$, at rate $1/(a\log(t/a))$ if primes near t/a have their expected local density uniformly in a (beyond what is known). Ignoring these difficulties, over $(t, t+h]$ the large-prime part changes by about $-h\sum_{a<t/\sqrt{v}}\mu(a)/(a\log(t/a)) = (1+o(1))h/\log^2 t$ by Lemma 11.3. Since $M(v) - M(u) = 0$, the $\sqrt{v}$-smooth part drifts at rate about $-1/\log^2 v$. The drift does not depend on σ_ν. For $\ell_\nu \gg (\log v_\nu)^4$ both parts exceed $\sqrt{\ell_\nu}$, and (11.2) is a balance between two opposite drifts: the local analogue of Proposition 10.15.

12. A model for the black curve

Status. Heuristic throughout.

Heuristic 12.1 (Scale and period)**.** This is a heuristic argument. Motivated by (5.1), one is led to model $M(x) \approx \sqrt{x}\,F(\log x)$ with F almost periodic, dominated at low frequency by $\gamma_1 = 14.1347\ldots$. Within an excursion of length $\asymp x$ this suggests $H_\nu \asymp \sqrt{x}$ and $\sqrt{H_\nu \ell_\nu} \asymp x^{5/4}$, so the black curve should oscillate at scale

$$|\mathcal{A}(x)| \asymp x^{5/4}, \tag{12.1}$$

with a period $2\pi/\gamma_1$ in $\log x$, that is, a factor $e^{2\pi/\gamma_1} \approx 1.56$ in x.

13. Numerical data

Status. Observations recorded in version 5; no new computation was carried out for version 6.

Experiment 13.1 (Excursion census)**.** *Object:* the excursion data $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$ and the curves $\mathcal{A}_+, \mathcal{A}_-, \mathcal{A}$. *Range:* $n \leq N = 455\,052\,511$, with $Z(N) = 87\,385$ excursions. *Expected outcome:* the bound (6.1); the prediction $Z(N)N^{-1/4} \approx 6.0 \times 10^2$ excursions longer than $\sqrt{N} \approx 2.13 \times 10^4$ from Heuristic 7.5. *Observed outcome (version 5):*

$$\mathcal{A}_+(N) = 6.3449 \times 10^9, \quad \mathcal{A}_-(N) = -7.0877 \times 10^9, \quad \mathcal{A}(N) = -7.4279 \times 10^8, \tag{13.1}$$

$$\max_\nu \frac{H_\nu}{\sqrt{\ell_\nu}} = 2.0743, \qquad \max_\nu \frac{\beta_\nu}{\ell_\nu^{3/2}} = 1.0214. \tag{13.2}$$

The count of excursions longer than $\sqrt{N}$ and the maxima restricted to $\ell_\nu > \sqrt{N}$ were not recorded. At this range $N^{0.05} \approx 2.7$, so factors x^ε are not separable from constants.

Warning 13.2 (Constant in the transfer bound). Version 5 compared $|\mathcal{A}(N)|$ with (6.1) using $c = \max_{10^3 \leq n \leq N} |M(n)|n^{-1/2}$, reporting a bound about 30 times larger than $|\mathcal{A}(N)|$. That c does not satisfy the hypothesis of Lemma 6.1, which requires all $n \leq x^*$; since $M(1) = 1$, any admissible c is ≥ 1. The comparison must either use the full maximum or split off the range below 10^3.

14. Open questions and future work

Status: OPEN / HEURISTIC ONLY. No item in this section is asserted as a theorem unless it explicitly cites an earlier theorem.

1. **OPEN, RH-EQUIVALENT.** Question 8.12: the uniform bilinear bound on square-ordinary excursion bridges. There is currently *no proved RH-scale bound* from the bilinear representation alone. By Corollary 8.6, proving the stated bound would prove RH.
2. **OPEN IN THIS NOTEBOOK.** Question 9.7: finiteness of square-exceptional excursions. KP86 does not settle it because at $a = 0$ it controls sign changes of $M + 2$, not zeros of M.
3. **OPEN.** Whether (6.4) is equivalent to RH. The former 5/6 zero-free deduction is *FALSE / WITHDRAWN.*
4. Whether the pruned local condition (7.8) holds, and a third bridge constraint (Remark 7.12).

(5) Experiment: the number of excursions with $\ell_\nu > \sqrt{N}$ against the prediction ≈ 600, and the maxima of $H_\nu/\sqrt{\ell_\nu}$ and $\beta_\nu/\ell_\nu^{3/2}$ over those excursions.
(6) Experiment: split each excursion by (11.1) and test Heuristic 11.4, in particular the independence of the drift from σ_ν.
(7) Experiment: frequency of touching runs (Definition 2.3) and of ballistic excursions (Heuristic 7.9).

Notebook History

Version 5. Single section: definitions, exact formulas, internal correlations, bounds, criteria, prime horizon, copy propagation, model, census data.

Version 6. Restructured into labelled sections following the authoring rules. Corrections: zero runs and touching zeros (Definitions 2.1, 2.3, Lemma 3.3); Remark 6.5; Remark 6.4; hypothesis of Lemma 6.1; constant in Remark 7.11; $j = 1$ in Proposition 9.6; (10.7); main term (Proposition 10.11, Warning 10.13). New: Warning 9.2, Lemma 9.4, Proposition 10.5, Proposition 10.15, Section 11, Heuristic 7.9, Corollary 7.10, proofs of Propositions 7.1 and 5.1. Archived as superseded: the calibration remarks (Warning 10.19) and the margin test (Warning 10.18).

Version 7. Full merge of version 6 with the reviewed changes to Sections 2–4. New exact material: terminal skeleton, g-skeletons, real/Hermitian triangular identity, extreme-lag identity, exact terminal quadratic decomposition, sharp (L, H) bounds and the 1/12 fixed-length constant. New central results: Lemma 8.2 ($O(\log\log X)$ power-exception count), Lemma 8.3 ($O(\log X)$ linear-long count), the verified Kaczorowski–Pintz large-oscillation input, and the unconditional localized RH criteria, in particular (8.7). The square range is written as the exact bilinear form (8.9); (8.10) records why prefix determinacy itself is not special to the square.

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